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SUPERSEDING
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6 April 1978

MILITARY SPECIFICATION

LENS ASSEMBLY, OBJECTIVE FOR NIGHT VISION GOGGLES, AN/PVS-5( )

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Lens Assembly, Objective, referred to herein as the objective, which is part of Night Vision Goggle, AN/PVS-5 ( ).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	- Pallet, Materials Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
QQ-S-781	- Strapping, Steel, Flat and Seals
NNN-P-40	- Paper, Lens
PPP-B-566	- Boxes, Folding, Paberboard
PPP-B-585	- Boxes, Wood, Wirebound
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-621	- Boxes, Wood, Nailed and Lock Corner
PPP-B-636	- Boxes, Shipping Fiberboard
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple Wall
PPP-S-760	- Strapping, Nonmetallic (and Connectors)
PPP-T-97	- Tape, Packaging/Industrial, Filament Reinforced

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of
MIL-B-117	- Bags, Sleeves and Tubing - Interior Packaging
MIL-C-675	- Coating of Glass Optical Elements
MIL-O-13830	- Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Cdr, USACERCOM, ATTN: DRSEL-RD-TS-S, Fort Monmouth, NJ 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-150	- Photographic Lenses
MIL-STD-171	- Finishing of Metal and Wood Surfaces
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-810	- Environmental Test Methods
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment

DRAWINGS

USA ELECTRONICS COMMAND

SM-D-657379	- Objective Lens Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Construction. The objective shall be in accordance with SM-D-657379 and all provisions of this specifications.

3.2 First article. The supplier shall furnish and test objectives as specified in the contract or purchase order (see 6.2).

3.3 Parts and materials. Parts and materials shall be as specified herein and as shown on applicable drawings. Materials not specified shall be selected by the contractor.

3.4 Performance characteristics.

3.4.1 Equivalent focal length. The equivalent focal length of the objective shall be 26.8 millimeters (mm), ±0.4 mm.

3.4.2 Flange focal distance. The flange focal distance of the objective cell assembly shall be 0.8055 ±.0002 inch where the measurement is taken from the front side of the chamfer on the front of the 1.124-inch diameter surface as shown on SM-D-657379.

3.4.3 Concentricity. The image displacement of the mechanical and optical axes of the objective shall not exceed 0.005 inch as shown on SM-D-657379.

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3.4.4 Linear distortion. The linear distortion of the objective at the edge of the 18 mm format shall be between 2.5 and 8.5 percent barrel distortion.

3.4.5 Field of view. The field of view of the objective shall be 40 degrees, +1 degree or -3 degrees.

3.4.6 F/number. The f/number of the objective shall be no greater than f/1.4.

3.4.7 T-number. The T-number of the objective shall be no greater than T-1.60.

3.4.8 Stray light. The stray light of the objective shall be no greater than 3.5 percent.

3.4.9 Modulation transfer function (MTF). The MTF of the objective shall be no less than that specified in Figures 1 and 2.

3.4.10 Field flatness/field tilt. The field flatness/field tilt of the objective lens shall be within 0.3 mm across the entire 18 mm format for radial and/or tangential focus.

3.4.11 Relative illumination. At full aperture, the image illuminance at the edge of the format shall be no less than 30 percent of the image illuminance on axis.

3.4.12 Torque. The objective lens assembly shall withstand a minimum torque of 10 foot-pounds (ft-lb) applied to the 8 start thread while the optical cell (see 6.3.5) is held stationary without damage or relative movement.

3.4.13 Weight. The weight of the objective shall not exceed 75 grams.

3.5 Environmental.

3.5.1 Vibration. The objective shall not be damaged (see 6.3.1) by vibrations in the frequency range of 10 to 55 hertz (Hz) at an amplitude level of 0.050 inch, ±0.005 inch double amplitude.

3.5.2 Temperature, humidity, altitude. The objective shall withstand storage at temperatures ranging from -57°C to +65°C, altitudes ranging from sea level to 50,000 feet above sea level while at a temperature of -57°C, and a humidity level up to 95 percent relative humidity at a temperature of +65°C. There shall be no damage, internal moisture, fogging, or frosting.

3.5.3 Temperature shock. The objective shall be capable of withstanding temperature changes from -57°C to +65°C in 5 minutes or less and from +65°C to -57°C in 5 minutes or less without developing any internal moisture, fogging or frosting and without damage.

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3.5.4 Immersion. The objective shall not be damaged and shall not show any evidence of water or moisture within the lens or leakage between the optical cell and any sleeve which may be affixed to it in fabricating SM-D-657379, after being immersed in fresh water to a depth of not less than 3 feet for a period of not less than 30 minutes.

3.6 Treatment and painting. Unless otherwise specified, the metallic parts of the objective shall be given a treatment and painting per MIL-STD-171 and optical surfaces shall be coated in accordance with MIL-C-675.

3.7 Marking. The objective shall be marked in accordance with SM-D-657379.

3.8 Workmanship. All objectives shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristics. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing. Objectives shall be free from burrs and shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material. The cleaning method shall not be injurious to any of the objectives, nor shall the parts be contaminated by the cleaning agent. Scratches and digs shall be as specified in the optical glass surface quality requirements and the cement defects of MIL-O-13830.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (4.3).
b. Quality conformance inspection (4.4).
c. Inspection for packaging (4.7).

4.3 First article inspection. Twenty-four objectives manufactured with production tooling shall be provided for first article and each shall be inspected in accordance with Tables I and II, subgroup I. In addition, 2 objectives selected at random shall be subjected to first article inspections in Table II, subgroup II. The tests may be performed in any order except that the Modulation Transfer Function shall be performed last. Failure of any test of Table I or Table II may be cause for rejection of first article. If molding is used for manufacture of optical elements, optical cells, or any part thereof, the first article samples shall be chosen so as to have equal representation from each and all molds.

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TABLE I. Inspections.

Categories	Defect	Method of Inspection
See SM-D-657379		
Critical	None Defined	
MAJOR		
101	1.1875-16 UNS-3A 8 start thread missing, improper or not concentric	Gage
102	1.0625-32 UNS-3A, thread missing or improper	Gage
103	1.124 diameter surface out of tolerance or concentricity	Gage/Measurement Optical
104	0.735 longitudinal dimension out tolerance	Gage/Measurement
MINOR		
201	Surface finish of 1.124 diameter surface not as specified	Visual/Gage
202	Marking, missing or not as specified	Visual
203	Packaging, missing or not as specified	Visual
204	Workmanship (see Para. 3.8)	Visual

TABLE II. First article tests.

Test	Requirement paragraph	Test paragraph
Subgroup I		
Equivalent focal length	3.4.1	4.5.1
Flange focal distance	3.4.2	4.5.2
Concentricity	3.4.3	4.5.3
Field of view	3.4.5	4.5.5
f/number	3.4.6	4.5.6
T/number	3.4.7	4.5.7
Stray light	3.4.8	4.5.8
Relative illumination	3.4.11	4.5.11
Torque	3.4.12	4.5.12
Weight	3.4.13	4.5.13
Vibration	3.5.1	4.6.1
Temperature, humidity, altitude	3.5.2	4.6.2
Temperature shock	3.5.3	4.6.3
Immersion	3.5.4	4.6.4
Modulation transfer function	3.4.9	4.5.9

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TABLE II. First article tests. (Cont)

Test	Requirement paragraph	Test paragraph
Subgroup II		
Linear distortion	3.4.4	4.5.4
Field flatness/field tilt	3.4.10	4.5.10

4.4 Quality conformance inspection.

4.4.1 Inspections. Objective shall be inspected for the attributes listed in Table I. Sampling for major defects shall be per MIL-STD-105, Level II, AQL 1 percent. Sampling for minor defects shall be per MIL-STD-105, Level II, AQL 4 percent.

4.4.2 Tests. For purposes of Group A, Group B and Group C testing a lot shall be defined as 50 objectives or one month's production, whichever occurs first.

4.4.2.1 Group A tests. Ten objectives from each lot shall be randomly selected and tested for the attributes listed in Table III. Failure of an objective to meet the requirements specified shall be cause for rejection of that objective. Failure of 2 or more objectives from the same lot, to meet the requirements specified shall be cause to revert to 100 percent inspection of the lot.

TABLE III. Group A tests.

Test	Requirement paragraph	Test paragraph
Flange focal distance	3.4.2	4.5.2
Concentricity	3.4.3	4.5.3

4.4.2.2 Group B tests. Lots which have passed the inspections of 4.4.2.1 shall be inspected in accordance with Table IV. Sampling for inspection shall be as described herein.

4.4.2.2.1 Sampling for Group B, subgroup I. One objective from each lot that has passed Group A tests shall be randomly selected and tested for the Group B, subgroup 1 attributes listed in Table IV. Failure to meet the requirements specified shall be cause for rejection of that objective and cause to revert to 100 percent inspection of the lot for the failed attribute until such time as 5 consecutive objectives have passed Group B, subgroup I tests.

4.4.2.2.2 Sampling for Group B, subgroup II. One objective shall be randomly selected from every third lot that has passed Group B, subgroup I inspections and tested for the Group B, subgroup II attributes listed in Table IV. Failure to meet the requirements specified shall be cause for rejection of that objective and cause to revert to 100 percent inspection of the lot for the failed attribute until such time as 5 consecutive objectives have passed Group B, subgroup II inspections.

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TABLE IV. Group B tests.

Test	Requirement paragraph	Test paragraph
Subgroup I		
Equivalent focal length	3.4.1	4.5.1
T-number	3.4.7	4.5.7
Straylight	3.4.8	4.5.8
Modulation transfer function	3.4.9	4.5.9
Torque	3.4.12	4.5.12
Subgroup II		
Linear distortion	3.4.4	4.5.4
Field of view	3.4.5	4.5.5
f/number	3.4.6	4.5.6
Field flatness/field tilt	3.4.10	4.5.10
Relative illumination	3.4.11	4.5.11
Weight	3.4.13	4.5.13

4.4.2.3 Group C tests. Lots which have passed the inspections of 4.4.2.2 shall be inspected in accordance with Table V. Sampling for inspection shall be as described herein.

4.4.2.3.1 Sampling for Group C, subgroup I. One objective shall be randomly selected from each 50 objectives produced and subjected to the inspection of Table V, subgroup I. Failure of any objective shall be cause for failure of Group C, subgroup I.

4.4.2.3.2 Sampling for Group C, subgroup II. One objective shall be randomly selected from each 50 produced during alternate months and subjected to the inspections of Table V, subgroup II. Testing shall be performed in the order listed in Table V. The first samples selected shall be from the second month production with subsequent samples selected every second month thereafter. Failure of any objective shall be cause for failure of Group C, subgroup II.

TABLE V. Group C inspections.

Test	Requirement paragraph	Test paragraph
Subgroup I		
Immersion	3.5.4	4.6.4
Subgroup II		
Vibration	3.5.1	4.6.1
Temperature, humidity, altitude	3.5.2	4.6.2
Temperature shock	3.5.3	4.6.3

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4.4.2.4 Actions required in the event of Group C failure. In the event of failure of the inspections of Group C, subgroup I, all objectives represented by the samples shall be subjected to the tests of subgroup I. Failure of any objective to pass the inspection shall be cause to reject that objective. In the event of failure of the inspections of Group C, subgroup II, actions as specified in the contract shall be taken (see 6.2).

4.5 Test procedure. Unless otherwise specified, all tests shall be performed at room temperature. Temperature chambers shall maintain specified temperatures within ±2°C. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment.

4.5.1 Equivalent focal length. The equivalent focal length of the objective shall be measured in accordance with the procedures of MIL-STD-150. Failure to meet the requirements of 3.4.1 shall constitute failure of this test.

4.5.2 Flange focal distance. The flange focal distance of the objective shall be measured in accordance with the procedures outlined in MIL-STD-150. Failure to meet the requirements of 3.4.2 shall constitute failure of this test.

4.5.3 Concentricity. Concentricity shall be measured in accordance with MIL-STD-150. Failure to meet the requirements of 3.4.3 shall constitute failure of this test.

4.5.4 Linear distortion. The linear distortion of the objective shall be measured in accordance with MIL-STD-150. Failure to meet the requirements of 3.4.4 shall constitute failure of this test.

4.5.5 Field of view. The field of view of the objective shall be measured in accordance with MIL-STD-150. Failure to meet the requirements of 3.4.5 shall constitute failure of this test.

4.5.6 f/number. The f/number of the objective shall be calculated by dividing the equivalent focal length by the effective aperture. The effective aperture of the objective shall be measured in accordance with the procedures of MIL-STD-150. Failure to meet the requirements of 3.4.6 shall constitute failure of this test.

4.5.7 T-number. The T-number of the objective shall be measured in accordance with the procedures of MIL-STD-150. The light source used shall be operated at 2856 K, ±200 K. The detector used shall have an S-20 extended red (ER) response complying with Figure 3. Failure to meet the requirements of 3.4.7 shall constitute failure of this test.

4.5.8 Stray light. The stray light of the objective shall be measured in accordance with Method 35, of MIL-STD-150. The detector used shall have an S-20 ER response complying with Figure 3. Failure to meet the requirements of 3.4.8 shall consititute failure of this test.

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4.5.9 Modulation transfer function. The objective shall be tested on a sine wave test pattern or equivalent set-up that utilizes an S-20 ER response photo-multiplier conforming to Figure 3 and a tungsten illumination source operated at 2856 K, ±200 K. If a microscope objective is used to relay the image of the objective, the numerical aperture (NA) of the microscope objective must be equal to or greater than 0.65 and have a magnification no greater than 20 power (X). The pattern-collimator combination must allow the objective lens to be tested either continuously or at a minimum of 8 distinct evenly distributed points from 0 lp/mm to 40 lp/mm with the first point being less than or equal to 1. Measure the MTF of the objective under test on axis. Move to the 14 degree field position without a change in focus and measure the radial and tangential MTF. Failure to meet requirements of 3.4.9 shall constitute failure of this test.

4.5.10 Field flatness/field tilt. Best focus shall be measured in the infinity focal plane of the lens (using a traveling microscope with an objective lens of numerical aperture (NA) equal to or greater than 0.65 and 20 power (20X) magnification). A resolution target (illuminated with a tungsten source operated at 2856 K, ±200 K, projected at infinity to the lens under test, shall be used to determine best focus positions. Readings shall be taken at a minimum of four evenly distributed, distinct points on each side of the optical axis, with the end points corresponding to the edge of the field for the lens under test. The lens shall be rotated 90 degrees and the measurements shall be repeated. The focus difference between the position of maximum best focus (radial and/or tangential and the position of minimum best focus (radial and/or tangential) shall be determined. Failure to meet the requirements of 3.4.10 shall constitute failure of this test.

4.5.11 Relative illumination. The relative illumination of the objective shall be measured in accordance with methods 7, 8, and 9 of MIL-STD-150. Failure to meet the requirements of 3.4.11 shall constitute failure of this test.

4.5.12 Torque. While holding the optical cell toward the front end apply a minimum torque of 10 foot-pounds (ft-lb) about the optical axis to the 8 start thread in a clockwise direction, then in a counterclockwise direction. Repeat this two additional times. Failure to meet the requirements of 3.4.12 shall constitute failure of this test.

4.5.13 Weight. The objective shall be placed on a suitable scale. Failure to meet the requirements of 3.4.13 shall constitute failure of this test.

4.6 Environmental. Each objective shall demonstrate its ability to perform properly after being subjected to all the Group C, subgroup II tests. For the purpose of post environmental testing, "perform properly" shall be defined as the ability to successfully complete in order the torque (4.5.12), modulation transfer function (4.5.9), stray light (4.5.8), and flange focal distance (4.5.2) tests.

4.6.1 Vibration. The objective shall be mounted to a test fixture and subjected to harmonic motion applied in each of three orthogonal planes; the frequency of the harmonic motion shall be varied uniformly from 10 to 55 Hz and return to 10 Hz in one minute; the amplitude shall be 0.050 inch, ±0.005 inch double amplitude; this frequency sweep shall be repeated 10 times in each plane. Failure to meet the requirements of 3.5.1 shall constitute failure of this test.

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4.6.2 Temperature, humidity, altitude. The objective shall be tested in accordance with method 518.1 of MIL-STD-810 except that in steps 3 and 13 the temperature shall be -57°C and the objective shall be inspected for the presence of internal moisture, fogging, or frosting at the completion of steps 3, 5, 7, 9, 13, and 14. Step 15 shall be limited to a visual check for damage. Lens cups may be used on the objective during this test. Failure to meet the requirements of 3.5.2 shall constitute failure of this test.

4.6.3 Temperature shock. The objective shall be tested in accordance with method 503.1 of MIL-STD-810 except that the high temperature extreme shall be +65°C and the objective shall be held at the temperature extremes for 2 hours in each step instead of 4 hours. Step 9 shall be limited to a visual check for damage. Failure to meet the requirements of 3.5.3 shall constitute failure of this test.

4.6.4 Immersion. The objective shall be subjected to Test Method 512.1, Procedure I of MIL-STD-810 except that the immersion time shall be 30 minutes minimum. Failure to meet the requirements of 3.5.4 shall constitute failure of the first half of the test. The objective shall then be placed in a fixture such that the imaging end is inside a watertight container which seals around the 1.124 inch diameter shoulder of the objective and the light input end is exposed to the environment and subjected to Test Method 512, Procedure I of MIL-STD-810 except that the immersion time shall be 30 minutes minimum. Failure to meet the requirements of 3.5.4 shall constitute failure of the second half of the test. Failure to pass either the first or second half of the test shall constitute failure of the test.

4.7 Inspection for packaging. Preparation for packaging shall be inspected in accordance with MIL-P-116 to determine compliance to requirements of Section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. The objective shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. The objective shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

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5.1.1.4.1 Technical literature. Each technical literature shall be unit packed method IC-1.

5.1.1.4.2 Lens assembly, objective. Each objective shall be individually unit packed method IA-8 as follows: wrap each unit with lens tissue conforming to NNN-P-40. Cushion each assembly by wrapping with cushioning material conforming to PPP-C-843, type II. Place the cushioned item within a barrier bag fabricated of material conforming to MIL-B-117, type II, class E, style 1. Heat seal the closure, leaving a small opening for exhausting air. Exhaust the entrapped air and complete the seal. Place the bagged item within a close-fitting box conforming to PPP-B-566, variety 2. Place the technical literature unit packed as specified in 5.1.1.4.1 on top of the contents directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.1.4.3 Intermediate packing. Multiples of 5 not to exceed 20 objectives unit packed as specified in 5.1 shall be placed within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B. Cleaning, drying, preservation application, unit packing and intermediate packing shall be as specified in 5.1, except that method III shall be used in lieu of method IA-8.

5.1.3 Commercial preservation. Preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B or Commercial, as specified (see 6.2). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of objectives, unit packed as specified in 5.1 shall be packed within a close-fitting box conforming to PPP-B-601, overseas type: PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.2.2 Level B. A quantity of objectives, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-640, class 2, style E, or PP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds.

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Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 43 x 24 inches or more and the width exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.3 Commercial packing. Packing shall be in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The objective is used with the AN/PVS-5( ) for general purpose viewing during periods of darkness.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Time frame required for submission of first article and number of assemblies required (see 3.2).
c. Level A, level B or Commercial preservation and packing (see section 5).
d. Necessary action by contractor in event of Group C failures.
e. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.
f. That data item DI-T-1903, DI-T-1906 and supplements thereto shall be a part of any contract incorporating this specification to guard against the use of materiel with hazardous levels of radioactive materiel.
g. For objectives being procured by the Government as an integral part of Night Vision Goggles, AN/PVS-5 ( ), the inspections of Table I, the Group A testing, the weight testing and the environmental testing per this specification shall be waived.

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6.3 Definitions.

6.3.1 Damage. Damage is defined as a failure of any finish, condensation, fogging or moisture on any internal surface; dents, or leakage of any hardware, connection or component.

6.3.2 Room temperature. Room temperature is defined as +23°C, ±10°C.

6.3.3 Inspection lot size. An inspection lot is defined as not more than those assemblies successfully completing Group A tests each month.

6.3.4 Environmental tests. For the purpose of this specification environmental tests are those tests discussed in paragraphs 4.6.

6.3.5 Optical cell. The optical cell is defined as the cylindrical part which is in direct contact with the lens elements, holds the lens elements in their fixed relative position, and has no external sleeves, flanges, or other surfaces or fittings attached to it in any manner.

Custodian:
Army - ER
Air Force - 11

Preparing activity:
Army - ER

Project No. 5855-0178

Review activity:
Army - AT
Air Force - 99

User activity:
Army - AT

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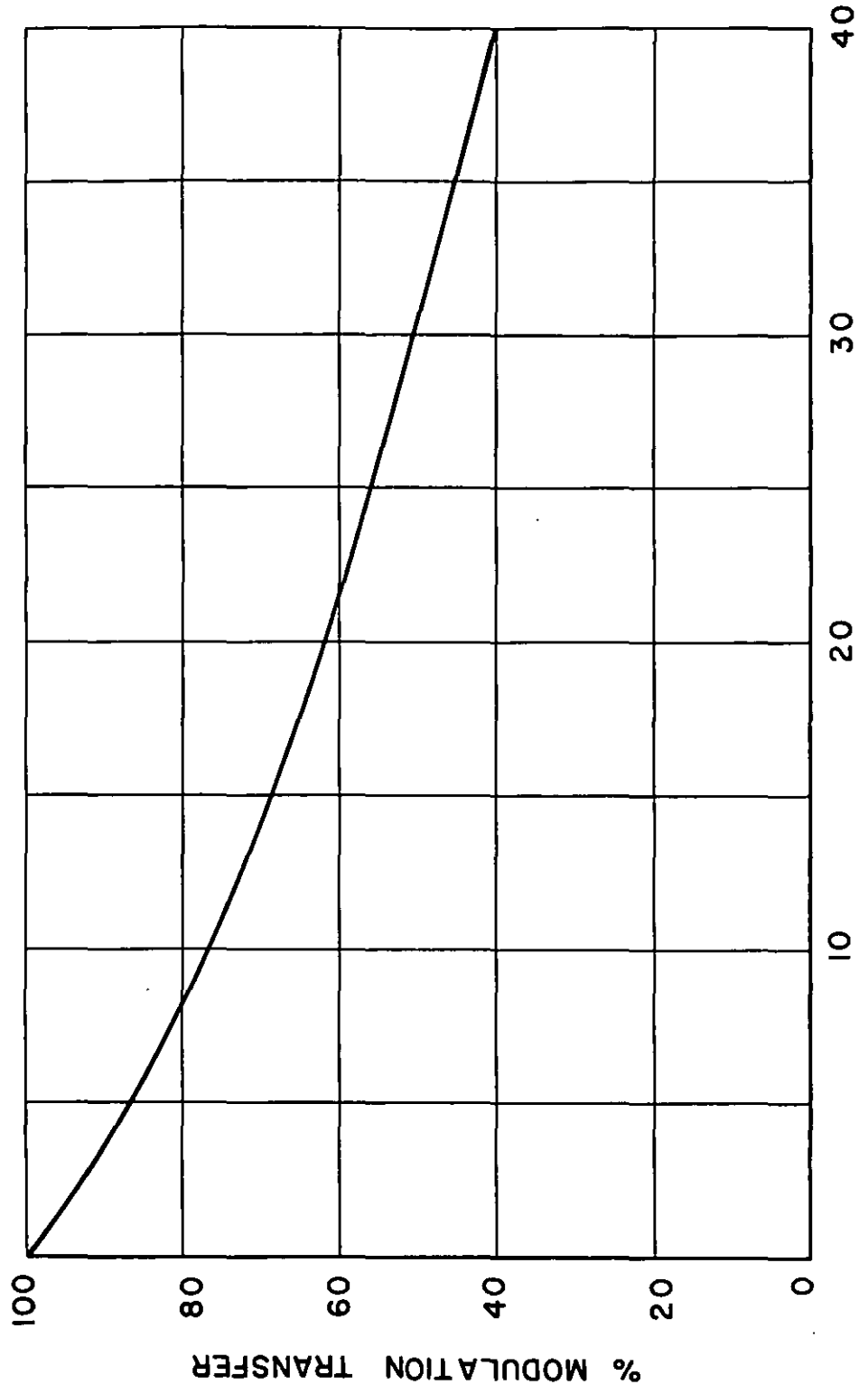


FIGURE 1. ON-AXIS MTF, OBJECTIVE

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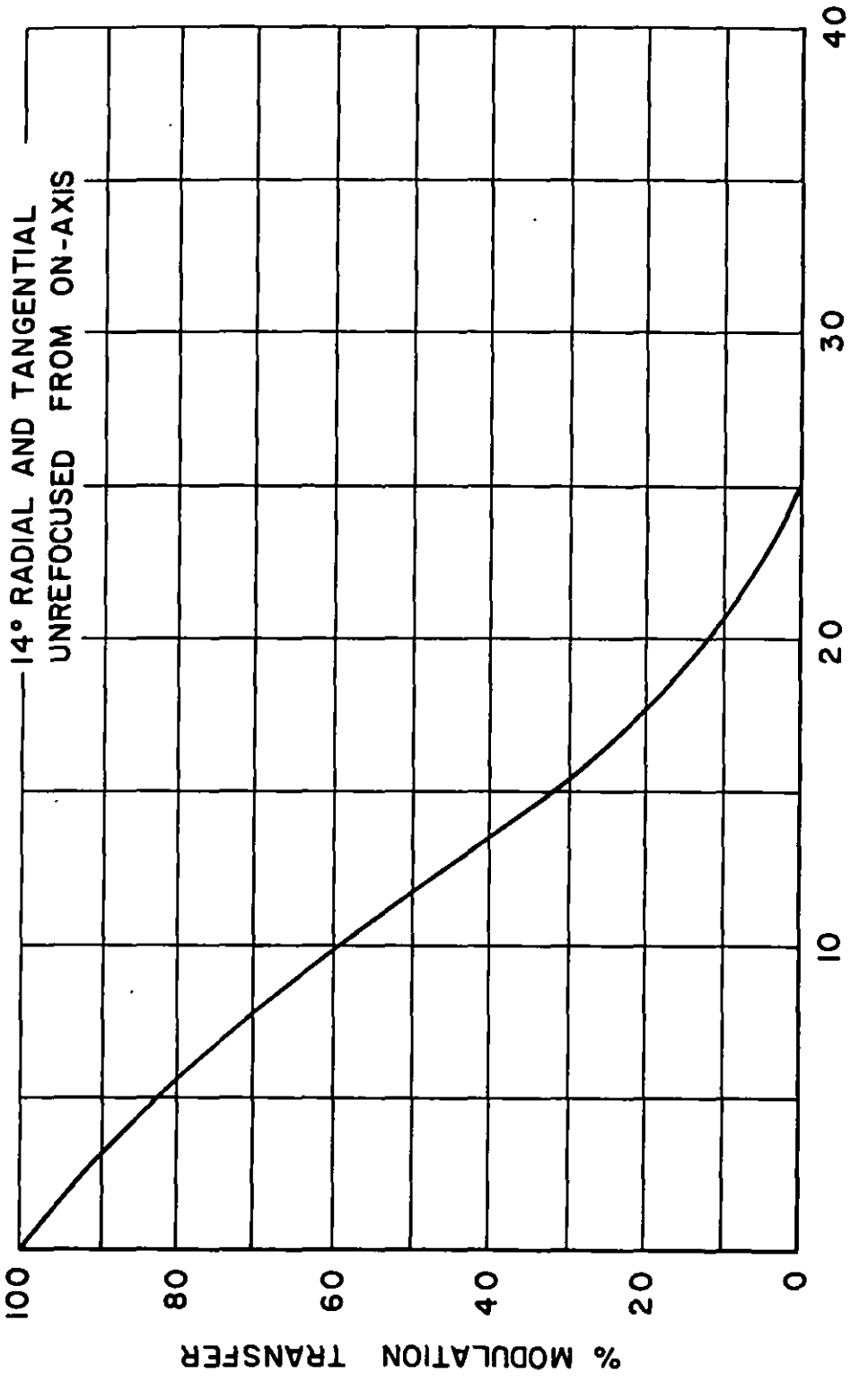


FIGURE 2. OFF-AXIS MTF, OBJECTIVE

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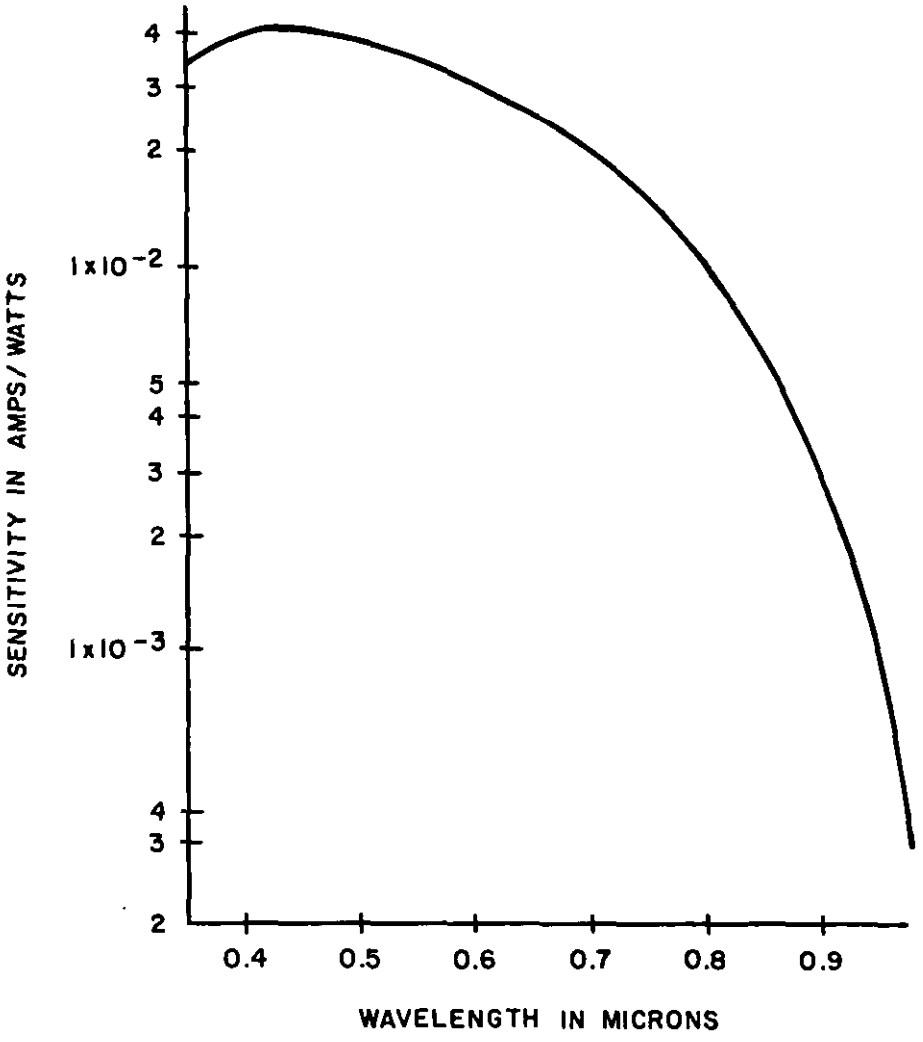


FIG. 3. S-20 EXTENDED RED RESPONSE CURVE

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